

1. Try Few-Shot first

Paste this stanza

Πάνω στην άμμο την ξανθή
γράψαμε τ' όνομά της·
ωραία που φύσηξεν ο μπάτης
και σβήστηκε η γραφή.

Giorgos Seferis, *Denial*, stanza 2.

Open Greek Rhyme System

Verification is on by default: untick it before this first run.

Identification settings

- Strategy: **Few-Shot**
- RAG: **OFF**
- Verification: **OFF**

Use your own API key or the key supplied for today. Keep the same model for every run.

Select **Ανάλυση** (Analyze).

Work in pairs. Save the answer.

2. “No need for symbolic rules?”

Did Few-Shot get these right?

Lines	Pair	Check for
1 and 4	ξανθή / γραφή	M, IDV
2 and 3	όνομά της / μπάτης	F2, MOS

If it found the mosaic rhyme...

“There you go. Why do we need symbolic verification?”

Would you trust it on another poem?

3. The answer was already in the prompt

Actual excerpt from the Few-Shot prompt

EXAMPLE 2 - Feminine-2 Mosaic:

Line 6: "γράψαμε τ' όνομά της" ['Gra-psa-me 'to-no-'ma tis]

Line 7: "ωραία που φύσηξεν ο μπάτης" [o-'re-a pu 'fi-si-ksen o 'ba-tis]

→ Classification: **F2-M0S**

[...]

- Rhyme spans word boundary in L6 → M0S

The same pair, its label and its explanation were supplied as a worked example.

Success here does not establish generalisation to an unseen example.

Source: greek-rhyme/prompts.py, FEW_SHOT, Example 2. Its line numbers refer to the original poem.

4. Now remove the worked examples

Same stanza, same model

Πάνω στην άμμο την ξανθή
γράψαμε τ' όνομά της·
ωραία που φύσηξεν ο μπάτης
και σβήστηκε η γραφή.

Select **Ανάλυση** again. Save this answer too.

Judge the returned answer against the stanza.

Change the strategy

- **Zero-Shot Structured**
- RAG: **OFF**
- Verification: **OFF**

This prompt gives the taxonomy and procedure, without the worked examples.

Inspect the difference

Does it still find **F2-MOS**?

Look for a missed mosaic label, a stress error or an invented feature.

5. Switch on phonological verification

Keep **Zero-Shot Structured**, the same model and **RAG OFF**.

Turn **Verification ON**, then select **Ανάλυση**.

The rule-based analysis

ξανθή / γραφή

M, IDV

όνομά της / μπάτης

F2, MOSAIC

The rhyme domain crosses the boundary in **μά | της**. The stressed vowel and following sounds match those in **μπάτης**.

A supplied example can guide an answer. A symbolic check supplies explicit evidence.

Read the three output sections

1. INITIAL LLM ANALYSIS

What does the model claim?

2. PHONOLOGICAL VERIFICATION

What do the rules find?

3. LLM REFLECTION & CORRECTION

Did the model use the report to correct its answer?

What changes when we make our assumptions explicit?

1. Theory in the prompt

Cairns-inspired emotion
scripts: situation, appraisal
and response.

2. Theory in rules

Four NeSy modes: inspect
an inference, a correction or
a disagreement.

3. Query the account

Zeugma: extract historical
claims, then ask what
follows.

<https://stergioscha.github.io/svarna/demos.html>

Stergios Chatzikyriakidis, ILSP | CLARIN:EL, 24 September 2026
Hands-on module. Saved outputs accompany the activities.

Achilles puts away the sword. What happens to his anger?

Homer, *Iliad* 1.172 to 1.224, A. T. Murray's English translation.

The threat

"I will myself come to your tent and take the fair-cheeked Briseis, your prize"

Achilles' response

"Grief came upon the son of Peleus"

"whether he should draw his sharp sword"

Athene intervenes

"But refrain, and obey us."

The ending

"and in no way ceased from his wrath"

Distinguish the emotion, the intended action and the completed action.

Use the complete passage in the appendix or `medea_cairns_passage_en.txt`.

Murray, 1924–1925, Perseus, public-domain translation. Quotations are excerpts.

Run the same passage with and without the Cairns prompt

Open Emotion Knowledge Graph

Keep these settings

Text Language: **English**

Analysis Mode: **Comprehensive**

AI Provider: **Auto**

Include Negative Emotions: **on**

Paste the complete Iliad passage. Keep the same text and settings for both runs.

1. Leave **Enable Script-Based Analysis** unticked.
2. Select **Analyze Emotions**. Inspect **Emotions**; save with **Download JSON**.
3. Tick **Enable Script-Based Analysis**, under **Enhanced Analysis (Cairns Framework)**.
4. Analyze and save again.

What does the second answer make explicit that the first leaves implicit?

Standard analysis: a plausible ending to question

Returned emotion labels

anger grief hatred wonder **relief**

The model's relationship claim

"Grief is resolved by relief as Athene intervenes and calms Achilles."

The passage

"But the son of Peleus again addressed with violent words the son of Atreus, and in no way ceased from his wrath:"

Does obedience establish relief? What evidence would justify that label?

Observed live API run: GPT-4o, 16 September 2026. One run per emotion configuration.

Cairns: theory changes the structure of the question

Actual opening of the enhanced prompt

“Analyze this text for emotional content using SCRIPT-BASED REPRESENTATION with METAPHOR DETECTION.”

Script component	Returned analysis of anger
------------------	----------------------------

Eliciting conditions	Insult to honor; threat to possessions
----------------------	--

Appraisals	“I have been disrespected”; “My status is challenged”
------------	---

Action tendencies	Confront; seek revenge; assert dominance
-------------------	--

Cultural norms	Anger is justified in defense of honor
----------------	--

The prompt also asks for bodily responses, expressive behaviour and moral evaluations.

Which connections can you defend from the text?

Observed enhanced output. Prompt excerpt from the inspected local template, `medea_cairns_prompt.txt`.
Cairns-inspired prompt from the inspected app source.

Reveal: prompt examples reappear as “textual evidence”

Phrase	Prompt example	Returned evidence	In passage
“Burning with rage”	yes	yes	no
“Filled with grief”	yes	yes	no
“Seized by anger”	yes	yes	no

A richer account still needs evidence

The output also supplies tears and increased heart rate.

A script’s typical features need not have occurred in this scene.

Mark each claim as quoted evidence, interpretation or unsupported addition.

Observed enhanced output, checked against the input and local prompt.

The overlap is consistent with prompt-example contamination; this is not a controlled causal experiment.

Four groups, four ways of using the theory

Open NeSy Emotions

Mode	Mechanism	Inspect
Cairns NeSy1	Heuristic rules and feedback	A rule and its proposed correction
Cairns NeSy2	pyDatalog rules and inference	The premises supporting an inference
Cairns NeSy3	Explicit emotion families and action concepts	What the vocabulary admits or misses
Cairns NeSy4	Model/grammar dialogue and a critic	A defended reading or proposed amendment

Same Iliad passage. **Genre Context: Epic; Language: English; AI Provider: Auto.**

Leave **Enable Feedback Loop** on. Select your mode, then **Analyze (Neuro-Symbolic)**.

One operator per group. Bring back one claim, its evidence and your verdict.

NeSy3 is selected by default. Choose the assigned mode explicitly. Save with Download JSON.

NeSy1: follow a correction all the way back to the text

1. The passage

"Grief came upon the son of Peleus"

2. The rule's objection

"Expected co-occurring 'fear' not found for 'grief'"

3. The feedback result

The entity labelled **grief** becomes **anger**.

The feedback item is marked **resolved**.

Observed trace

10 entries

6 fired rules

4 inconsistency records

4 feedback calls

The feedback discusses Agamemnon, although the target grief belongs to Achilles.

Was a missing emotion found, or did the feedback replace a supported label?

Observed live API run: GPT-4o, 16 September 2026. One run per emotion configuration.

NeSy2: inspect the premise as well as the inference

Rule or check in the returned trace	Observed result
Anger alongside grief	Dependency satisfied
Anger deepens grief	Causal rule fired
Grief requires co-occurring fear	Missing dependency
Anger requires a hubris-family item	Missing dependency

After two feedback calls

The grief entity is again relabelled **anger**.

Four trace entries, two fired rules, two inconsistency records.

Does grief without fear contradict this passage, or challenge the rule?

Observed live API run: GPT-4o, 16 September 2026. One run per emotion configuration.

NeSy3: what does a clean grammar result establish?

Returned labels

wrath
grief
indignation
helplessness

Returned checks

Grammar compliance: **1.0**
Rule firings: **0**
Inconsistencies: **0**
Symbolic corrections: **0**

The passage explicitly says: “Achilles was seized with **wonder**”.
Wonder is missing from this constrained extraction.

What can fall outside the vocabulary while everything inside it passes?

Observed live API run: GPT-4o, 16 September 2026. One run per emotion configuration.

NeSy4: the model can challenge the grammar

The unconstrained reading adds wonder

“Achilles was seized with wonder, and turned around, and immediately recognized Pallas Athene.”

Dialogue: defend

“The passage explicitly states that Achilles was ‘seized with wonder,’ which directly supports the proposed emotion of wonder”

Proposed ontology amendment

Type: **new family**

Proposed name: **wonder**

Evidence: the quoted passage

A proposal for human review. It was not applied.

Would you extend the grammar, revise the label, or defend its existing scope?

Observed NeSy4 output. Defence quoted in part. Six dialogue turns, all defend; four amendment proposals.

NeSy4: fitting the vocabulary can also change the reading

Open **Dialogue (NeSy4)** and compare the free reading with the critic.

Care becomes pity

Passage: Hera “loved and cared for both men alike”.

Free reading: **care**

Critic revision: **pity**

Obedience becomes submission

Passage: Achilles obeys while remaining angry.

Free reading: **obedience**

Critic revision: **submission**

Assigned family: **helplessness**

Do these revisions preserve what you want to say about the scene?

The main four-emotion result remains the baseline. Dialogue, critic revisions and amendment proposals are separate outputs.

Observed live API run: GPT-4o, 16 September 2026. One run per emotion configuration.

Zeugma: extract the Theban entry into Plataea

Open Zeugma

Thucydides 2.2

Theban troops enter Plataea at night.

Naucleides opens the gates.

The intruders make a proclamation, hoping the city will join them.

Scene summary. The verbatim selection and queries are together in the appendix; the saved graph follows this slide.

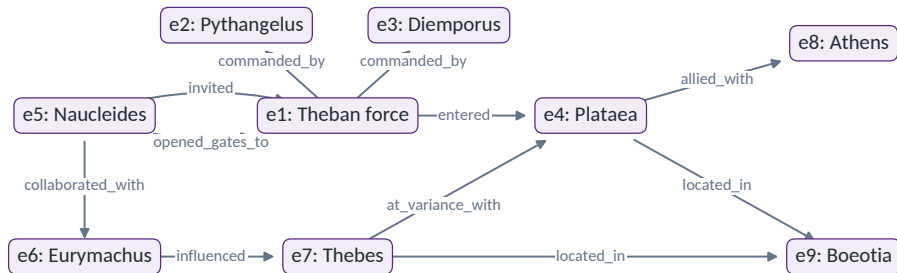
1. Paste the text under **Step 1: Ancient Text for Graph Extraction**.
2. Keep **Reasoning Mode: Basic - General logical reasoning** and the same provider.
3. Select **Step 1: Extract Knowledge Graph**.
4. Inspect **Available Predicates**.

Find one actor, one place and one relationship that the passage supports.

One shared extraction for the room. Source text: medea_zeugma_passage_short_en.txt.
Richard Crawley translation, Perseus Digital Library, Tufts University. CC BY-SA 4.0.

The saved graph used for the six queries

Shorter Thucydides selection: **9 entities, 11 relations.**



The diagram preserves the returned `allied_with` edge. The successful query uses the literal property `alliance(e4, 'Athens')`.

Drawn from the saved API response, 16 September. Every returned entity and relation is shown.
Exact facts, source text and query answers: `zeugma_materials.pdf`. IDs belong to this extraction.

Ask a question, then follow the shared variable

Under **Step 2: Zeugma Symbolic Queries**, enter one complete query per line.

Select **Run Queries**.

Who commanded the force?

```
commanded_by(Force, Commander),  
entity(Commander, Name, person)
```

Observed: **Pythangelus** and **Diemporus**.

Who opened the gates to a force entering which city?

```
opened_gates_to(Person, Force), entered(Force, Place),  
entity(Person, Name, person), entity(Place, City, place)
```

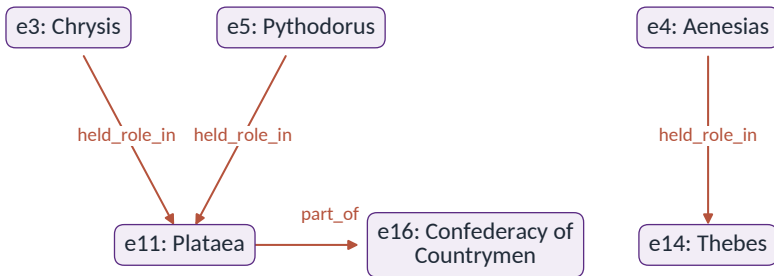
Observed: **Naucleides**, **Plataea**.

The repeated variable connects the facts. Are those facts supported?

Queries wrap visually here; paste each query as one line. Names and IDs may change after a fresh extraction.

The saved graph behind the historical-error discussion

Separate run: full Thucydides 2.2. These are its four disputed edges.



Read the arrows as the model's assertions. Compare them with the source text on the next slide.

Exact IDs and predicates from the saved full-passage response. This is a selected subgraph.
The complete 16-entity, 14-relation graph is in the appendix and `zeugma_materials.pdf`.

The graph turns an invitation into an established fact

Saved extraction from the **full** Thucydides 2.2 passage.

An invitation becomes membership

Text: The herald invites Plataea to join the confederacy.
The Thebans hope it will join.

Graph: Plataea **is part of** the confederacy (part_of).

The text never reports acceptance.

Other errors: office locations

Person	Text	Graph
Chrysis	Argos	Plataea
Aenesias	Sparta	Thebes
Pythodorus	Athens	Plataea
Graph predicate: held_role_in.		

Why the check misses these errors: it checks that the linked entities exist.

It reports consistency **1.0** and zero contradictions without checking textual support.

Correct reasoning over this graph can still give a wrong historical answer.

Observed full-passage extraction. The current UI does not run the separate paper's historical validation workflow.

When a query returns nothing, check what failed

The full passage

The apostrophe in **Thirty Years' Truce** broke the generated Prolog.

Seven of eight queries returned no solutions.

The page had reported successful extraction.

The shorter passage

Five of six initial queries worked.

An alliance rule failed; the direct **alliance** property worked.

All six classroom queries returned answers for the saved graph.

A missing fact, a wrong representation, or an execution error?

Use the tested selection and inspect **Detailed Output Log**. Keep the source passage beside the results.

Both returned RDF exports failed a Turtle syntax check.

Bring back one claim you can defend

Prompt

Which theoretical distinction made the reading more precise?

Which added detail lacked evidence?

Rules

Which premise supported the inference?

Which “correction” would you reject?

Queries

Which answer followed from the graph?

What must still be checked against the source?

Your contribution: a claim, its textual evidence, and the assumption it depends on.

The apps remain available after the session: <https://stergioscha.github.io/svarna/demos.html>

Funding from Microsoft's LINGUA project supported CoDAG (Computational Dialectal Atlas of Greek), making Svarna, GPT-4.1 fine-tuning and the Greek NLP Swiss Knife possible.

Nature Analysis: interpretation through an LLM

Open Nature Analysis

What the model supplies

Flora, fauna, sea, land and weather.

Metaphor, personification, emotional tone and attitudes towards nature.

What the application adds

Segmentation, filtering, counts and exports.

No independent symbolic verifier.

What counts as nature: a physical setting, a metaphor, or a memory?

Choose the original teaching passage or the Papadiamantis story introduced below.

Whole Text tested live with Claude Sonnet 5, 20 September 2026. Model output still needs interpretation.

Run the Greek passage paragraph by paragraph

Input and settings

File: nature_excerpt.txt

Segmentation: **Paragraphs**

Minimum words: **20**

Nature threshold: **0.2**

Keep all five paragraphs. The full text is reproduced in the appendix.

Before running it, decide what you would count in each paragraph.

Original fictional passage written for this exercise, not attributed to a published author.
Whole Text tested with Claude Sonnet 5. Compare paragraph results during the exercise.

1. Use the existing login and your own or supplied provider key.
2. Paste under **Paste your text here:**.
3. Select the provider and model, then **Analyze Nature Content**.
4. Save with **Export JSON** or **Export CSV**.

Five paragraphs, five annotation decisions

Part	Evidence in the text	Decision to discuss
1	Coast, trees, gull, wind and rain	Which elements are literally present?
2	Office, application and contract	Does the surrounding landscape affect this label?
3	“θύελλα αντιδράσεων”, “βαθιές ρίζες”	Are metaphors counted as weather or flora?
4	Angry sea, absent reeds, remembered frogs	Mention, present entity or remembered environment?
5	Chapel, olive trees, labour and memory	Does a religious gesture establish a religious attitude towards nature?

Declare the annotation policy before judging the model.

Questions grounded in the teaching passage. These are not observed model errors.

Change the unit, then inspect the statistic

Save the first result. Change only **Segmentation** to **Whole Text** and rerun.

What the nature percentage counts

Successful segments above the nature threshold, divided by all successfully analysed segments.

Paragraphs

Several separate judgements.

The administrative paragraph can differ from its neighbours.

Whole Text

One judgement for the entire passage.

The percentage now concerns a single segment.

Did the interpretation change, the counting unit change, or both?

Definition checked in the inspected source. Scores are LLM judgements; they are not independently verified.
Record the model and compare the two returned results.

Alternative: Papdiamantis, The Seal's Lament

Αλέξανδρος Παπαδιαμάντης

Τὸ Μυρολόγι τῆς φώκιας

A grandmother washes by the sea. Her granddaughter drowns unheard. A seal finds the child and the narrator gives its lament human words.

Full Greek text: papdiamantis_seal.txt

What to inspect

Landscape and animal life.

Human lament, flute and sea sounds.

The seal as animal and mourner.

Seaweed and shells as funeral adornment.

Φύκια ἔναι τὰ στεφάνια της,
κοχύλια τὰ προικιά της...

Can one tone label capture mourning, pleasure and the animal's feeding?

Society for Papdiamantis Studies: complete Greek text. First published 1908.

For this story use Paragraphs, Minimum words 5. The recorded test used a 500-word selection in Whole Text mode.

Voyant-NLP: can we find the seal?

Open Voyant-NLP

Paste `papdiamantis_seal.txt` under **Upload**; select **Add to Corpus**, then **Concordance**.

Search	Matches	Form in the text	What changed?
φώκια	0		Modern spelling misses the historical form.
φώκ	3	φώκη, φώκης	A shared fragment finds inflected forms.
φωκ	1	φωκῶν	The accent falls elsewhere.

Read the contexts: when is the seal feeding, listening or mourning?

Four explicit word-form matches do not recover every reference to the animal.

Observed in the live browser, 19 September 2026. Literal substring search; no LLM key needed.

Voyant-NLP: what does a zero sentiment score mean?

Open **Sentiment & Clusters**, then select **Analyze** under **Sentiment Analysis**.

Observed on the complete story

Average polarity: **0.0**

Average subjectivity: **0.0**

Every returned segment also scored zero.

Read the text beside the score

Bereavement, drowning and lament.

Music, pleasure and the animal's meal.

Which of these does polarity measure?

Does this result establish that the story is emotionally neutral?

The inspected source uses default TextBlob sentiment, an English-oriented analyzer, on Greek text.
Corpus statistics and NLP models require language and task checks too.

Open Svarna Greek Corpus Workbench

A separate app for existing corpora. Keep the database on **corpus**.

1. Open **Concordancer**; search **λοιπόν**.
2. Select **europarl** under **Corpus**, then search.
3. Repeat with **opensubtitles**; read beyond the first short rows.

Actual examples

Συμφωνούμε, λοιπόν. Παραδεχτείτε το λοιπόν!

Λοιπόν; Λοιπόν, λοιπόν, λοιπόν.

Inference, transition, an interactional prompt, or insufficient context?

A parliamentary source does not guarantee an inferential function. Subtitles are not spontaneous conversation.

Svarna dialectal: the same form across varieties

Change the database selector to **dialectal**. Search **ίντα**.

Corpus: grdd_cretan

ίντα μιλείς;

και δεν κατέχω, ίντα να πω

The second quotation is a fragment of a longer verse line.

Corpus: grdd_cypriot

Ίντα κλαίεις;

Ίντα μμάθκια!!

Compare the surrounding words and the punctuation.

A direct question, an embedded question, or an exclamation?

Does one Modern Greek gloss work in every example? Separate dialect, genre and spelling.

Both browser searches succeeded, 19 September 2026. A short bridge to the later dialect session.
Collection labels describe the source corpus; they do not verify every word's dialect.

Svarna LLM: classify a function or explain a pattern

Open **LLM**. Select a provider and enter a valid key for that provider.

Marker Classification

Enter **λοιπόν**.

Select **Ταξινόμηση** (Classification).

Up to 50 retrieved sentences.

Function, confidence and a Greek explanation for each sentence.

Pattern Synthesis

Enter **ίντα**.

Select **Ανάλυση** (Analysis).

Up to 30 sentences plus frequency data.

An English account of usage, position, functions and variation.

Label examples yourself, then check the model's claims against them.

Confidence is self-reported. No symbolic checker verifies the linguistic interpretation.

Inspect the evidence that the LLM actually receives

Observed browser input for the ίντα synthesis

Database: **dialectal**. Concordancer filter: **grdd_cypriot**.

Proposed LLM sample: **13 Cretan and 17 Cypriot rows**.

- The database choice carries through; the individual dialect filter does not.
- Sentence text is sent without its individual corpus label.
- The frequency table counts matching sentence rows, not every occurrence.

Which claims about dialect differences can this input support?

Treat the result as an interpretation of a mixed sample. Return to the source rows before accepting a generalisation.

Input captured locally before model submission; it is not a generated result.

Source-based prompt templates: `svarna_llm_prompts.md`.

PlotAnalyzer: let a theory guide the reading

Open PlotAnalyzer

Papdiamantis: The Seal's Lament

Complete Greek story:

`papdiamantis_seal.txt`

The child drowns. The grandmother misreads the sound. The reader knows more. The seal laments.

1. Select **Short Story**, then **Aristotelian (Poetics)**.
2. Enable **NeSy**, choose one model and configure its OpenRouter key.
3. Turn **Heuristic**, **Contrastive** and **Feedback Loop** off.
4. **Analyze Plot**, read the explanations, then **Export**.

Whose error counts as a flaw? Who recognizes the drowning?

With Feedback Loop off, the symbolic rules still check the model's estimates.

One dependency in the app's Aristotelian schema

“Suffering needs Flaw: the error must precede its consequences”

pathos (suffering) depends on **hamartia** (flaw).

1. Read the LLM's scores, reasons and positions. Locate the evidence in the story.
2. Open **Symbolic Reasoning Trace**: which rule adjusted which score?
3. Separate a character's discovery from the reader's knowledge.

If suffering is clear but a flaw is not, should we revise the reading or question the dependency?

The quotation is the app's operational rule, not Aristotle's words.

Inspect whether it fires in this run; its presence in the schema does not guarantee a violation.

PlotAnalyzer: send the objection back to the model

Same Greek text, theory and model. Turn **Feedback Loop** on.

LLM reading → rule check → targeted question
→ model reassessment → second rule check

1. **Analyze Plot**, then **Export** this run separately.
2. In **Symbolic Feedback Loop**, follow an objection, the model's response and the score change.
3. Read **Symbolic Reasoning Trace**. Check the revised reading against the text.

Did the response recover evidence, or simply fit the schema more closely?

Each run begins with a fresh model call. Follow a correction within its own feedback trace.
If there are no detected inconsistencies, there may be no feedback call.

PlotAnalyzer: change the lens, keep the story

Select **Russian Formalism**. Keep the input, model and engine settings fixed.

Aristotelian poetics

Flaw, reversal, recognition, suffering.

Whose action brings the reversal? Who comes to understand it?

Russian Formalism

Fabula and syuzhet, framing, defamiliarization.

How are simultaneous events presented? What changes when the seal speaks a human lament?

Which explanation helps us read this story? Which assumptions need defending?

Compare evidence and reasons. Overall scores use different feature sets across theories.
Complete Greek text, English synopsis and both theory prompts are in the HTML companion.

TermGuard: what is a cut?

Our excavation record distinguishes the **interface left by digging** from the **material that fills it**.

Candidate definition of cut

- A a cut made during excavation
 - B an interface created by the removal of material
 - C a deposit that occupies a pit
-

Which repeats the term? Which confuses the concept with a related one?

A category and a distinguishing characteristic help us inspect a definition.

We still have to ask whether the category is correct.

Instructor-written discussion examples, not model outputs or parser results.

Full input: termguard_stratigraphy.txt, original English teaching corpus, not a published standard.

TermGuard: upload, extract, then compare

Open TermGuard

1. **Project Management:** own project name; Domain: Archaeological stratigraphy. Click **Create**.
2. **Corpus Building:** upload `termguard_stratigraphy.txt`, Language: English. Click **Upload**.
3. Configure an authorised key in **API Keys**. Choose an available model, click **Extract Terms**, and inspect the returned rows.
4. In **Neuro-Symbolic Feedback Loop**, select **cut**, Max Iterations: **2**. Run these comparisons with the same model:

Zero-shot / No RAG → Persona (ISO) / No RAG

Persona (ISO) / No RAG → Persona (ISO) / Keyword RAG

Compare each definition with the corpus and the intended concept.

TermGuard: put the definition theory in the prompt

Zero-shot, complete template

Find the definition for the following term:

Domain: {domain}

Term: {term}

Output only the definition text.

Persona (ISO), excerpt

“An intensional definition states:

- the immediate generic concept, and
- the delimiting characteristics that distinguish the concept from others in the same category.”

The full prompt adds substitution and worked earthquake and mouse examples.

What did the theory add? What would the corpus add?

Separate parser checks can then produce feedback for a revision.

Templates retrieved from the live project's settings. Full text: `termguard_prompts.md`.

The app's ISO-inspired implementation. The template is called Few-shot in settings and Persona (ISO) in the dropdown.

Linguistic Distance: how different is Greek?

Open Linguistic Distance

1. Under **Language Pairs**, click **Clear All**; select only **Ancient Greek** → **Modern Greek**.
2. Under **Analysis Dimensions**, click **Select All**.
3. Click **Run Analysis**. Initialization is automatic.
4. Read the table, then choose **Create Visualization** or **Export Results**.

Which matters for your question: word forms, grammatical features or historical cognates?

This app calculates distances from linguistic resources.

No chat prompt or API key is needed.

Missing data remain N/A. Compare the available dimensions separately.

Greek: seven selected, six measured

Dimension	Distance	Evidence being compared
Lexical	0.5713	Retained word-list forms
WALS	0.2685	Coded structural features
Grambank	0.2371	Grammatical feature values
URIEL	0.1997	Resource-derived feature vectors
Syntactic	0.3230	Features extracted from UD treebanks
Phonological	N/A	No score returned for the pair
Cognate	0.4235	Cognate-set overlap across concepts
Combined	0.3372	Mean of the six available scores

N/A does not mean that pronunciation stayed the same.

Two live browser runs returned these same values, 19 September 2026. Rounded as in the UI.

These are app outputs, not published benchmark findings or a percentage of the language that changed.

Look behind the score: what was compared?

- **Lexical:** 136 retained word pairs from 200 positions.
Are meanings aligned? What do spelling conventions change?
- **Grambank:** 46 differing values among 194 compared features.
Which distinctions do these features represent?
- **Syntax:** 5,000 Ancient Greek and 4,324 Modern Greek sentences.
Check genre, period, sampling and annotation before explaining a difference.
- **Cognates:** 98 shared matches across 170 compared concepts.
Related words can still differ substantially in form.

The denominators differ. Are the scores answering the same question?

Counts from the actual analysis log. UD files: `grc_perseus` and `el_gdt`.

Inspect dataset editions, lexical alignment and preprocessing before interpreting historical change.

Question the average, keep the evidence

One number depends on choices

The combined score gives the available dimensions equal weight.

A missing dimension changes what that average represents.

- Which dimensions would you keep for your historical question?
- Do equal numerical scales make the evidence comparable?
- Calculate a new average from selected columns in the saved CSV. Explain what your choice leaves out.

Report the profile, its coverage and your selection of evidence.

No mutual-intelligibility or rate-of-change claim follows from this score.

Dialect Generator: one prompt, two dialect choices

Open Dialect Generator

Choose the existing example

Πες μου για το χωριό σου

1. Choose **Cretan** and **Llama 3.1, 8B Instruct**.
2. Set **Tokens: 40**; keep **Temp: 0.75**, **Top-p: 0.90**.
3. Click **Generate**. Copy the result and its settings.
4. After completion, select **Pontic** and repeat with the same prompt and settings.

One shared model and adapter; the dialect choice changes the prompt.

CPU generation needs no API key.

Compare the generated forms with examples from the corresponding corpus.

Dialect labels: examine the returned text

Cretan selection, excerpt of actual output

γιά-ε κειδέ! ετούτονέ ναι τ όνομά μου, κι έρχομαι από την βενετία.

Pontic selection, full output

ονοματίεις από φαίνεται πολλά καλόν χωρίον έν εγώ πάω κόρ αγοράζω και σο
μαγαζίν τσ εξέρ

Check completeness at this short token limit. The full Cretan output contains stray control characters.

Which forms can you support? Does the whole answer make sense?

Completed CPU streams: Cretan 312.43 seconds; Pontic 315.70 seconds, as reported by the app.

Wording preserved. Max Tokens was 40; no finish reason is reported. Assess the dialectal features in the returned text.

Return to attested language in Svarna

Open the separate Svarna Corpus Workbench

1. Choose **dialectal**, then corpus **grdd_pontic**.
2. Search **ατό**, a form in the generated text. Read its context and source label.
3. Record one supported observation and one unresolved question.

Actual returned Pontic corpus examples

Σωστόν έν ατό!

Το χρέος ατ ατό έτον κι ατό εποίκεν.

A corpus match supports an individual form. Assess the generated sentence separately, and consider possible training-data overlap.

Browser search completed, 19 September 2026. Corpus rows labelled Pontic, GRDD+. Corpus attestation alone does not establish an independent test.

Bring observations to Markantonatou, Bompolas, Stamou and Dimakis at 11:30.

Choose from the app bank

Research question	Prepared activities
Text and interpretation	Rhyme, MEDEA, NATS, Nature Analysis, PlotAnalyzer
Attested usage	Voyant-NLP and the separate Svarna Corpus Workbench
Specialist comparisons	TermGuard, Linguistic Distance, Dialect Generator

Choose modules within the 45-minute hands-on slot.

Inspect the text, run an analysis, then ask what supports its claims.

Keep saved outputs available when a live operation is slow or fails.

Funding acknowledgment

Funding from Microsoft's LINGUA project supported CoDAG (Computational Dialectal Atlas of Greek), making Svarna, GPT-4.1 fine-tuning and the Greek NLP Swiss Knife possible.

NATS remains a separate prepared activity. Additional functions inside these apps are optional extensions.

Source passage: Iliad 1.172 to 1.224, 1 of 3

Then the king of men, Agamemnon, answered him:

“Flee then, if your heart urges you; I do not beg you to remain for my sake. With me are others who will honour me, and above all Zeus, the lord of counsel. Most hateful to me are you of all the kings that Zeus nurtures, for always strife is dear to you, and wars and battles. If you are very strong, it was a god, I think, who gave you this gift. Go home with your ships and your companions and lord it over the Myrmidons; for you I care not, nor take heed of your wrath. But I will threaten you thus: as Phoebus Apollo takes from me the daughter of Chryses, her with my ship and my companions I will send back, but I will myself come to your tent and take the fair-cheeked Briseis, your prize, so that you will understand how much mightier I am than you, and another may shrink from declaring himself my equal and likening himself to me to my face.”

Homer, *Iliad*. A. T. Murray, 1924–1925, Perseus, public-domain translation.
Read all three pages as one continuous passage.

Source passage: Iliad 1.172 to 1.224, 2 of 3

So he spoke. Grief came upon the son of Peleus, and within his shaggy breast his heart was divided, whether he should draw his sharp sword from beside his thigh, and break up the assembly, and slay the son of Atreus, or stay his anger and curb his spirit. While he pondered this in mind and heart, and was drawing from its sheath his great sword, Athene came from heaven. The white-armed goddess Hera had sent her forth, for in her heart she loved and cared for both men alike. She stood behind him, and seized the son of Peleus by his fair hair, appearing to him alone. No one of the others saw her. Achilles was seized with wonder, and turned around, and immediately recognized Pallas Athene. Terribly her eyes shone. Then he addressed her with winged words, and said:

“Why now, daughter of aegis-bearing Zeus, have you come? Is it so that you might see the arrogance of Agamemnon, son of Atreus? One thing I will tell you, and I think this will be brought to pass: through his own excessive pride shall he presently lose his life.”

Homer, *Iliad*. A. T. Murray, 1924–1925, Perseus, public-domain translation.
Read all three pages as one continuous passage.

Source passage: Iliad 1.172 to 1.224, 3 of 3

Him then the goddess, bright-eyed Athene, answered:

“I have come from heaven to stay your anger, if you will obey, The goddess white-armed Hera sent me forth, for in her heart she loves and cares for both of you. But come, cease from strife, and do not grasp the sword with your hand. With words indeed taunt him, telling him how it shall be. For thus will I speak, and this thing shall truly be brought to pass. Hereafter three times as many glorious gifts shall be yours on account of this arrogance. But refrain, and obey us.”

In answer to her spoke swift-footed Achilles:

“It is necessary, goddess, to observe the words of you two, however angered a man be in his heart, for is it better so. Whoever obeys the gods, to him do they gladly give ear.”

He spoke, and stayed his heavy hand on the silver hilt, and back into its sheath thrust the great sword, and did not disobey the word of Athene. She returned to Olympus to the palace of aegis-bearing Zeus, to join the company of the other gods. But the son of Peleus again addressed with violent words the son of Atreus, and in no way ceased from his wrath:

Homer, *Iliad*. A. T. Murray, 1924–1925, Perseus, public-domain translation.

Read all three pages as one continuous passage.

a Theban force a little over three hundred strong, under the command of their Boeotarchs, Pythangelus, son of Phyleides, and Diemporus, son of Onetorides, about the first watch of the night, made an armed entry into Plataea, a town of Boeotia in alliance with Athens. The gates were opened to them by a Plataean called Naucleides, who, with his party, had invited them in, meaning to put to death the citizens of the opposite party, bring over the city to Thebes, and thus obtain power for themselves. This was arranged through Eurymachus, son of Leontiades, a person of great influence at Thebes. For Plataea had always been at variance with Thebes; and the latter, foreseeing that war was at hand, wished to surprise her old enemy in time of peace, before hostilities had actually broken out.

Richard Crawley translation, Perseus Digital Library, Tufts University. CC BY-SA 4.0.

Verbatim selection beginning within the opening sentence; chronology omitted. Join both pages for the tested input.

Indeed this was how they got in so easily without being observed, as no guard had been posted. After the soldiers had grounded arms in the marketplace, those who had invited them in wished them to set to work at once and go to their enemies' houses. This, however, the Thebans refused to do, but determined to make a conciliatory proclamation, and if possible to come to a friendly understanding with the citizens. Their herald accordingly invited any who wished to resume their old place in the confederacy of their countrymen to ground arms with them, for they thought that in this way the city would readily join them.

Richard Crawley translation, Perseus Digital Library, Tufts University. CC BY-SA 4.0.

Verbatim selection beginning within the opening sentence; chronology omitted. Join both pages for the tested input. Queries follow on the next two slides.

Thucydides 2.2: Prolog queries, 1 of 2

1. Commanders

```
commanded_by(Force, Commander), entity(Commander, Name, person)
```

Pythangelus; Diemporus.

2. Gate-opening and entry

```
opened_gates_to(Person, Force), entered(Force, Place),  
entity(Person, Name, person), entity(Place, City, place)
```

Naucleides; Plataea.

3. Alliance property

```
alliance(Place, AllyName), entity(Place, City, place)
```

Plataea; Athens.

Enter one query per line. These predicates belong to the saved short graph. Check Available Predicates after a fresh extraction.

Thucydides 2.2: Prolog queries, 2 of 2

4. Collaboration

```
collaborated_with(Person, Partner),  
entity(Person, Name, person), entity(Partner, PartnerName, person)
```

Naucleides; Eurymachus.

5. Places in a region

```
located_in(Place, Region), entity(Place, City, place),  
entity(Region, RegionName, region)
```

Plataea and Thebes; Boeotia. Thebes' location is background knowledge, not stated in this selection.

6. An asserted opposition

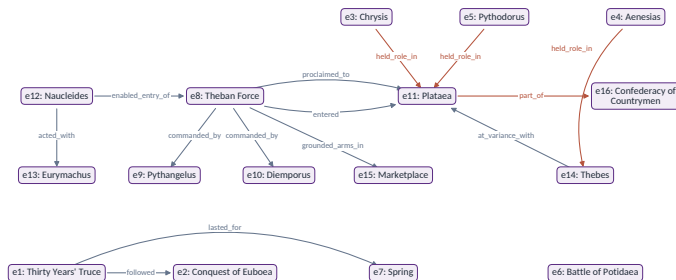
```
at_variance_with(e7, e4)
```

True in the saved graph: **e7 = Thebes**, **e4 = Plataea**. Check IDs after re-extraction.

Observed shorter-passage query results. Logical success concerns the stored representation.

Reference: the complete full-passage graph

Full Thucydides 2.2: **16 entities, 14 relations**. IDs differ from the shorter run.



Orange: the four disputed edges. Other assertions are preserved without certifying them. The isolated Battle of Potidaea entity is retained.

Drawn from the complete returned entity/relation arrays, 16 September.

Text, graph, all IDs, literal properties, queries and answers are together in `zeugma_materials.pdf`.

Nature exercise: full Greek passage, 1 of 3

Όταν η Ελένη επέστρεψε στο χωριό, ο δρόμος προς το λιμάνι μύριζε βρεγμένο χώμα. Τα αλμυρίκια έγερναν πάνω από την άμμο και ένας γλάρος στεκόταν στην άκρη του μόλου. Ο νοτιάς έφερνε μικρά κύματα ως τα σκαλοπάτια των σπιτιών. Πίσω από τις αυλές, τα πεύκα κρατούσαν ακόμη τις σταγόνες της νυχτερινής βροχής. Τίποτε δεν έμοιαζε έκτακτο, κι όμως εκείνη σταμάτησε σαν να έβλεπε τον τόπο πρώτη φορά.

Στο κοινοτικό γραφείο, η υπάλληλος της ζήτησε την ταυτότητά της και τον αριθμό της αίτησης. Η Ελένη περίμενε ώσπου να τελειώσει ένα τηλεφώνημα. Έπειτα παρέλαβε δύο αντίγραφα της απόφασης, υπέγραψε στο σχετικό βιβλίο και ρώτησε πότε είχε εγκριθεί η σύμβαση. Η υπάλληλος της έδειξε την ημερομηνία. Το όνομα του πατέρα της δεν υπήρχε στον κατάλογο των ιδιοκτητών, παρόλο που εμφανιζόταν στο παλαιότερο συμβόλαιο.

Original fictional text written for the workshop. Read all three pages as one passage, keeping the paragraph breaks.

Copyable file: nature_excerpt.txt. Whole Text tested with Claude Sonnet 5, 20 September 2026.

Nature exercise: full Greek passage, 2 of 3

Στο καφενείο, η απόφαση είχε σηκώσει θύελλα αντιδράσεων. Ένα κύμα διαμαρτυρίας περνούσε από τραπέζι σε τραπέζι, αλλά κανείς δεν συμφωνούσε για το επόμενο βήμα. Ο αδελφός της έλεγε πως το πρόβλημα είχε βαθιές ρίζες και πως οι υποσχέσεις του προέδρου ήταν μόνο καπνός. Εκείνη άκουγε χωρίς να μιλά. Όταν κάποιος τη ρώτησε αν θα προσέφευγε στο δικαστήριο, απάντησε ότι ήθελε πρώτα να διαβάσει τα έγγραφα.

Το απόγευμα κατέβηκε ξανά στην ακτή. Η θάλασσα την περίμενε θυμωμένη και χτυπούσε επίμονα την κλειστή πόρτα μιας αποθήκης. Στο ρέμα δεν υπήρχαν πια καλάμια. Ο πατέρας της έγραφε στο ημερολόγιό του ότι τις ανοιξιάτικες νύχτες δεν μπορούσε να κοιμηθεί από τα βατράχια. Τώρα ακουγόταν μόνο η αντλία του εργοταξίου. Η Ελένη έψαξε με το βλέμμα τις όχθες, χωρίς να ξέρει αν περίμενε να βρει κάτι ή να επιβεβαιώσει την απουσία του.

Original fictional text written for the workshop. Read all three pages as one passage, keeping the paragraph breaks.

Copyable file: nature_excerpt.txt. Whole Text tested with Claude Sonnet 5, 20 September 2026.

Από το ξωκλήσι ακούστηκε η καμπάνα και μια ηλικιωμένη γυναίκα σταυροκοπήθηκε. Λίγο πιο πέρα, ένας εργάτης έδενε κόκκινες κορδέλες στους κορμούς των ελιών που θα κόβονταν για να περάσει ο νέος δρόμος. Η Ελένη θυμήθηκε τη μητέρα της να απλώνει τα δίχτυα για τη συγκομιδή και να παραπονιέται για τη μικρή σοδειά. Για την ίδια, εκείνα τα δέντρα ήταν μαζί δουλειά, περιουσία και μνήμη. Δεν ήξερε ποια από τις τρεις λέξεις θα χωρούσε στην ένσταση.

Original fictional text written for the workshop. Read all three pages as one passage, keeping the paragraph breaks.
Copyable file: nature_excerpt.txt. Whole Text tested with Claude Sonnet 5, 20 September 2026.